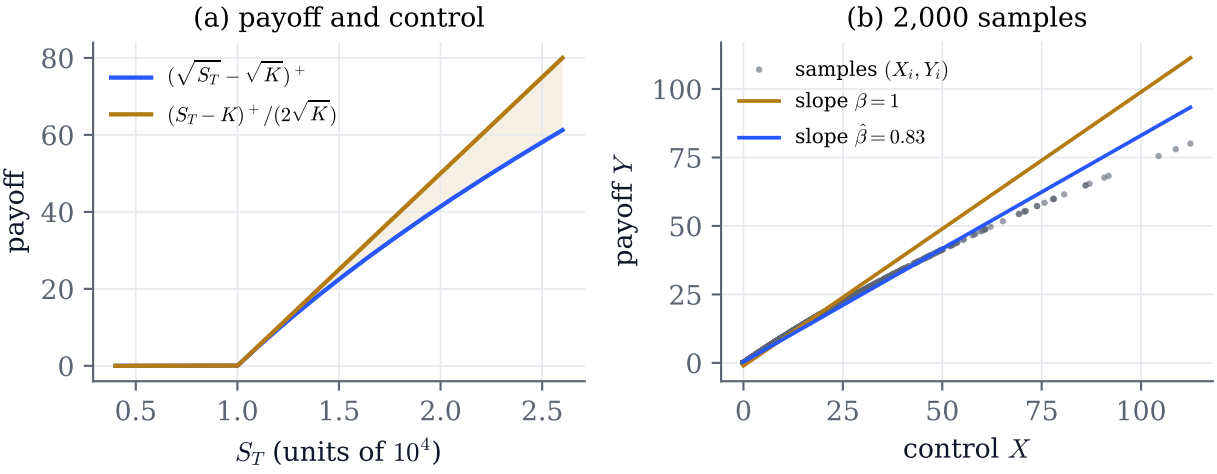

(a) payoff and control
payoff
80
60
40
20
0
$(\sqrt{S_T} - \sqrt{K})^+$
$(S_T - K)^+ / (2\sqrt{K})$
0.5
1.0
1.5
2.0
2.5
S_T (units of 10^4)
(b) 2,000 samples
payoff Y
100
75
50
25
0
samples (X_i, Y_i)
slope $\beta = 1$
slope $\hat{\beta} = 0.83$
0
25
50
75
100
control X